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Project Proposal: Ai Health and Fitness Agent

Problem Statement and Motivation: Many people want to lose fat, build muscle, or improve their health, but struggle to stay consistent because fitness plans often do not match their real-life constraints. Online advice is frequently too generic or too extreme and ignores factors such as limited time, equipment access, food preferences, and recovery. As a result, even motivated users may stall or burn out.

Personal trainers and nutrition coaches can help, but they are expensive and inaccessible to many students and young professionals. Most fitness apps rely on static templates and do not adapt well when users get busy or plateau. This project proposes an AI Health and Fitness Planning Agent that creates personalized training and nutrition plans, reviews them for safety and realism, and updates them using a multi-agent workflow.

Target Users: The system targets students and young professionals who want structured guidance for training and nutrition but do not have access to a coach. It is especially useful for users with limited time, limited equipment, or specific dietary preferences. The system is intended for general wellness planning but does not provide medical advice.

Why an Agentic AI Solution Makes Sense: Fitness planning involves balancing goals, schedules, training load, nutrition, and recovery. These elements influence each other, and conflicts can easily occur. A single chatbot response can miss issues such as pairing up a very low-calorie diet with excessive training volume. Using multiple agents allows each part of the problem to be handled separately. One agent focuses on training, another on nutrition, another checks for safety concerns, and a final agent combines all recommendations into a realistic plan that fits the user’s lifestyle.

Use Cases and Scenarios

Scenario 1: A user wants to lose fat over six weeks while maintaining strength and trains four days per week. The system builds a plan and adjusts it if it is too aggressive.

Scenario 2: A user wants to build muscle but can only train three days per week for 30 to 40 minutes. The system creates a time-efficient workout and nutrition plan.

Scenario 3: A user experiences a two-week plateau. The system identifies likely causes and suggests a small adjustment for the following week.

In each scenario, the system first gathers the user's goals and constraints, then generates training and nutrition plans, checks them for safety and realism, and finally combines everything into one clear plan with a short weekly check-in.

Initial System Design:
The system includes several specialized agents. The Intake Agent gathers user goals, schedule, experience, equipment access, and dietary preferences. The Training Plan Agent creates the workout structure. The Nutrition Plan Agent calculates calories and macronutrient targets. The Recovery and Safety Critic Agent checks for unsafe or unrealistic recommendations. The Plan Integrator Agent combines all outputs into a final plan.

The workflow begins with user input through a front-end interface. Training and nutrition plans are generated, reviewed by the critical agent, and finalized by the integrator agent which reads the full output (plan, schedule, meal plan)

Tools, APIs, and Data Sources:
The project will use a web-based front-end (thinking of using streamlit) for user input and plan that will be displayed. Internal tools such as a rule-based macro calculator and a local exercise library will be used to improve consistency and reduce unrealistic outputs. All tools and data sources will be free and publicly available.

Feasibility and Risks:
A key risk is generating unsafe or unrealistic advice. This is mitigated through rule-based constraints and a dedicated critic agent. Another risk is unclear user input, which will be addressed through structured intake forms. The system will clearly state that it provides general wellness guidance and not medical advice.

Timeline Aligned with Course Schedule and Constraint

Weeks 1–3 will focus on defining the project's scope and submitting the proposal by February 4.

Weeks 4–7 will focus on building core agent logic and supporting tools.

By Week 8, Milestone I will deliver a working front-end.

Weeks 9–10 will complete Milestone II with a full multi-agent workflow.

By Week 11, Milestone III will complete evaluation and pilot results.

Weeks 12–14 will be used for final implementation and presentation preparation.

Expected Outcome: The outcome will be a working agentic system that demonstrates how a multi-agent approach can produce safer, more realistic, and more useful fitness plans than a single-agent solution. The system will be evaluated based on how well it respects user constraints, identifies issues, and supports users in a small pilot test.